Phased Gradient Ultra Open Metamaterials for Broadband Acoustic Silencing

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Supplementary Note 1: Acoustic properties of super unit cells

Firstly, we would like to demonstrate that the acoustic properties of a single super unit cell with a hard boundary on both sides are equal to those of a model consisting of two super unit cells arranged in reverse. We conduct simulations on the pressure field distributions of various models, as shown in Fig. S1. Model 1 represents a single super unit cell with hard boundaries on both sides, model 2 represents two super unit cells arranged in reverse with hard boundaries on both sides, and model 3 represents two super unit cells arranged periodically with hard boundaries on both sides. To note that unit cells with the same number represent same units. Comparing the pressure field distributions of the three models, the pressure field distribution of model 1 is same as that of model 2 and different from that of model 3. This finding implies that model 1 can be regarded as equivalent to model 2 instead of model 3. Therefore, the acoustic properties of the PGUOM composed of a single super unit cell are not supposed to be the same as PGUOM multiple super unit cells.

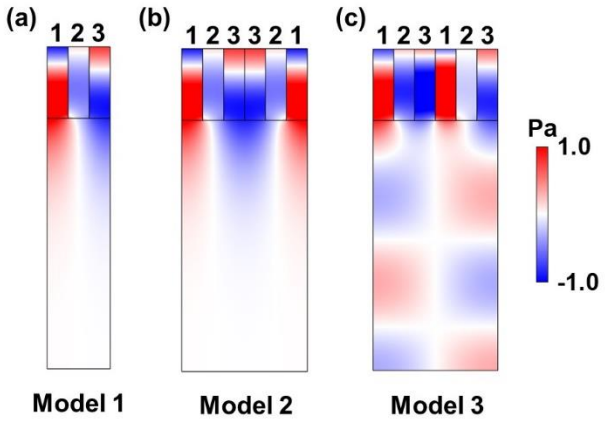


Fig. S1. Pressure field distribution of PGUOMs in different arrangements. (a) Pressure field distribution of a super unit cell with hard boundaries on both sides. (b) Pressure field distribution of two super unit cells arranged in reverse. (c) Pressure field distribution of two super unit cells arranged periodically.

Supplementary Note 2: Effect of amplitude and width of unit cell on the transmission of a PGUOM

We present transmission color maps of varied amplitudes for the PGUOM with unevenly distributed unit cells, as shown in Fig. S2. In this design, the phases of the unit cells form a phase gradient, while the amplitude and the width of each unit cell differ. We use "dL" to represent the width differences between unit cell or unit cell 3 and unit cell 2. The blue area indicates a transmission is close to 0, signifying a better silencing effect, while the red area denotes a weaker silencing effect. According to Fig. S2, when dL is small, a better noise reduction occurs when the amplitudes of the unit cells are close. Conversely, when dL is large, a better noise reduction occurs when the amplitudes of the shorter-length unit cell significantly exceed those of the longer-length unit cells.

In analysis section, we set the amplitudes of the three units to 1, however, in the validation section, the usage of length-varying straight barriers revealed variations in amplitude within each unit cell. Therefore, we investigated the impact of amplitude variations on the transmission of the PGUOM. In Fig. S3(a-f), we compared the pressure field distribution for different amplitudes of the transmission wave within each unit cell, noting the amplitudes for the three-unit cells at the corner of each figure. Remarkably, when the amplitudes of the side unit cells differ (as seen in Fig. S3(a, c, e)), the silencing effect is less pronounced. Conversely, when the amplitudes of the side unit cells are uniform (as shown in Fig.S3(b, d, f)), the silencing effect is more effective.

Another difference between the analysis section and the validation section is the frequency of the transmission dip. Our PGUOM targeted a frequency of 2000 Hz, the observed frequencies of dips were at 1950 Hz and 2150 Hz. To address this, we examined the pressure field distribution at frequencies of 1950 Hz, 2000 Hz, and 2150 Hz, as shown in Fig. S3(g-i). The pressure field distributions at 1950 Hz and 2150 Hz resembled the ideal pressure field distribution in Fig. S3(b), suggesting that the mechanism of the PGUOM remains effective. The frequency shift observed may be attributed to the inaccuracies in the height of the waveguide.

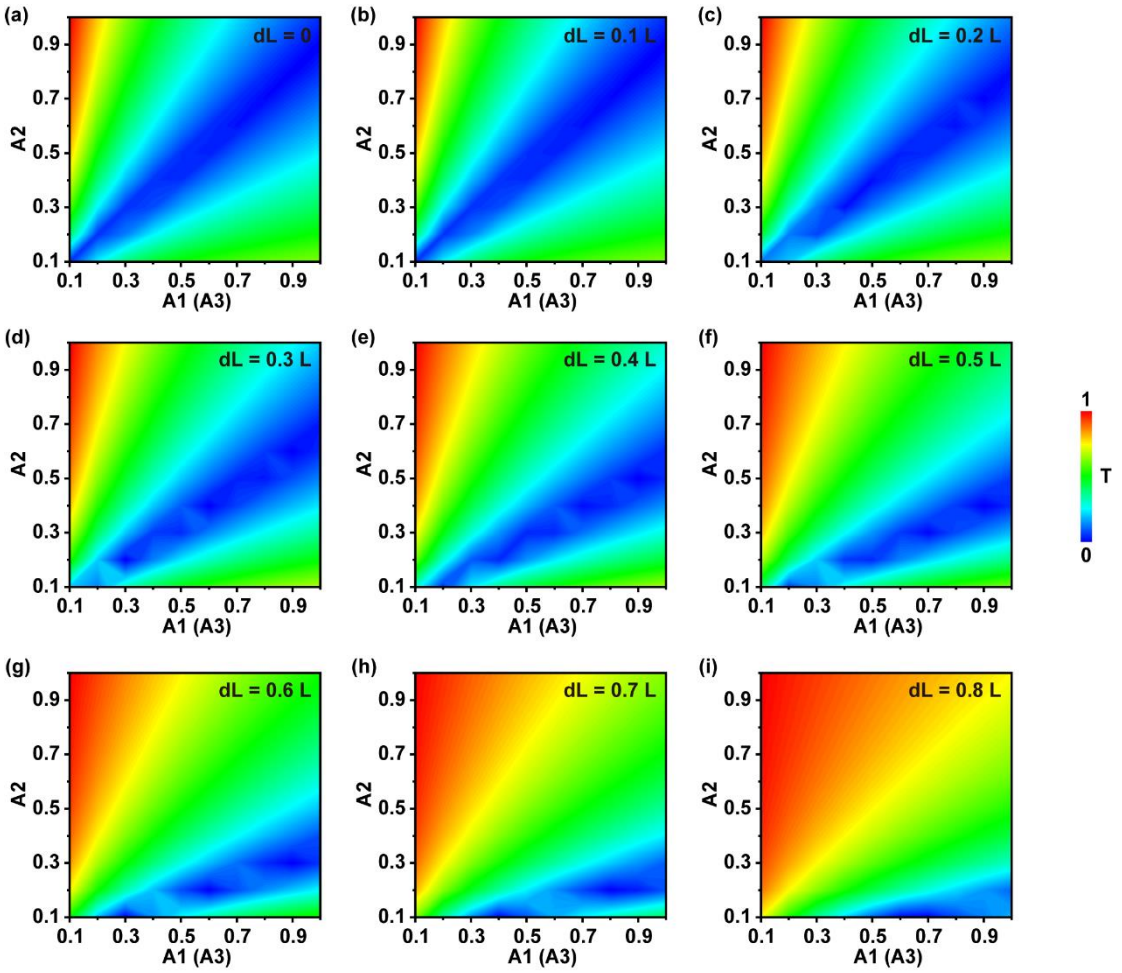


Fig. S2. Transmission coefficient as a function of the amplitudes of each unit cell of PGUOM. The amplitude of center units (A2) and side units (A1 and A3), ranging from 0.1 to 1, while the phase gradient is maintained. Length differences between the center units and side units are set ranging from 0 to 0.8.

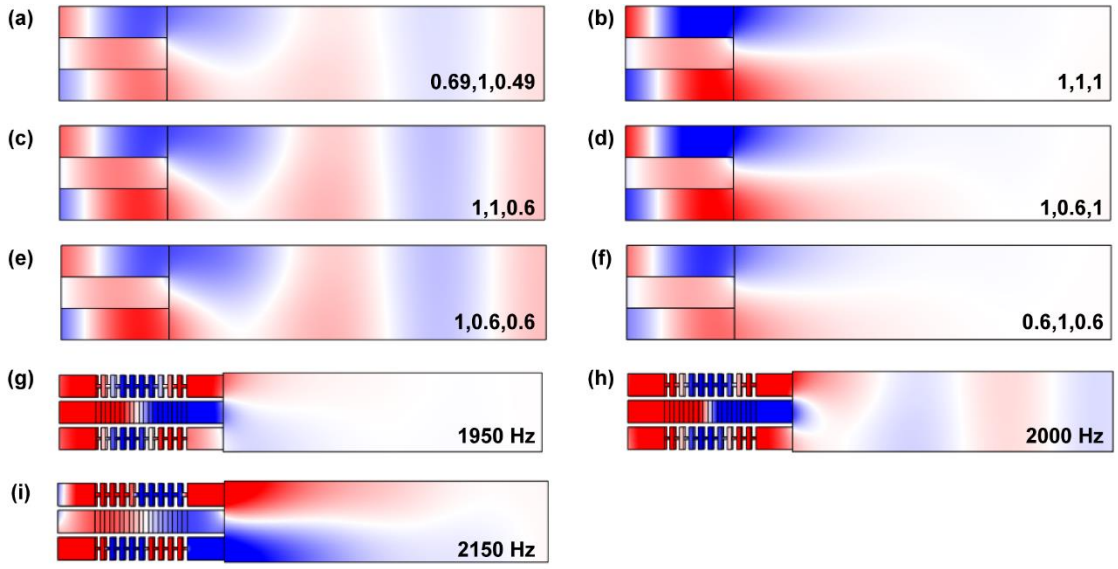


Fig. S3. Pressure field distributions of the theoretical and validated PGUOM. Pressure field distributions of the theoretical PGUOM, with amplitudes of the three-unit cells indicated in the lower right corner. (g-i) Pressure field distributions of the validated PGUOM at specific frequencies: 1950 Hz, 2000 Hz, and 2150 Hz.

Supplementary Note 3: Cylindrical PGUOM with a target frequency at 2000 Hz

To demonstrate the target frequency of the cylindrical PGUOM can be tuned to 2000 Hz with a similar outer radius of the design, we exhibit the simulation results of the PGUOM with various patterns in Fig. S4 and Fig. S5. Fig. S4 uses the same pattern as we used in the paper, fan shape slits for both unit cell 1 and unit cell 3. Fig. S4 uses a different pattern, a circular shape slit is used for unit cell 1. Both patterns reduce the fabrication resolution requirements. The design process is similar as we did in the paper: we first adjust the degree/ radius of unit cell 1 and unit cell 3 of the PGUOM to achieve a phase distribution spanning 2π with a similar amplitude, as shown in Fig. S4(b-c) and Fig. S5(b-c). Then, waveguides are added to achieve the transmission dip at the target frequency. Corresponding pressure field distributions and transmission spectra are plotted in Fig. S4(d-f) and Fig. S5(d-f). The simulation results demonstrate that the target frequency of the design is tunable, and the pattern can be optimized if the fabrication resolution cannot be satisfied.

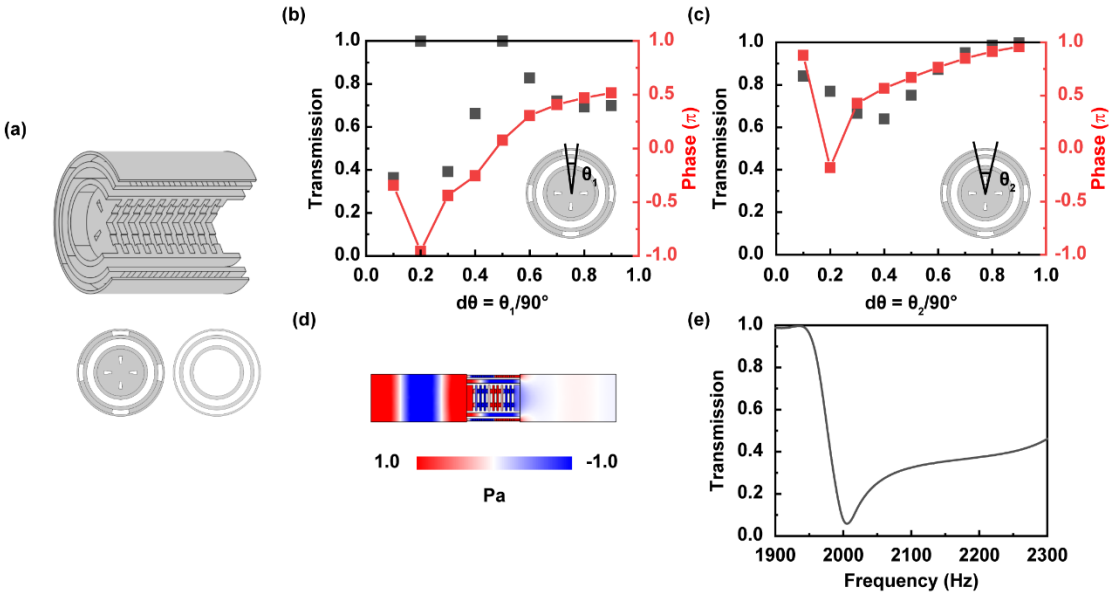


Fig. S4. Illustration and simulation results for cylindrical PGUOM (Design 1). (a) Illustration of the cylindrical PGUOM and its cross-section pattern. (b-c) Amplitude and phase of the transmission wave for the inner and outer layer at 2000 Hz, varying in angles. (d) Pressure field distribution of the cylindrical PGUOM at 2000 Hz. (e) Transmission spectrum of the cylindrical PGUOM across the frequency ranging from 1900 Hz to 2300 Hz.

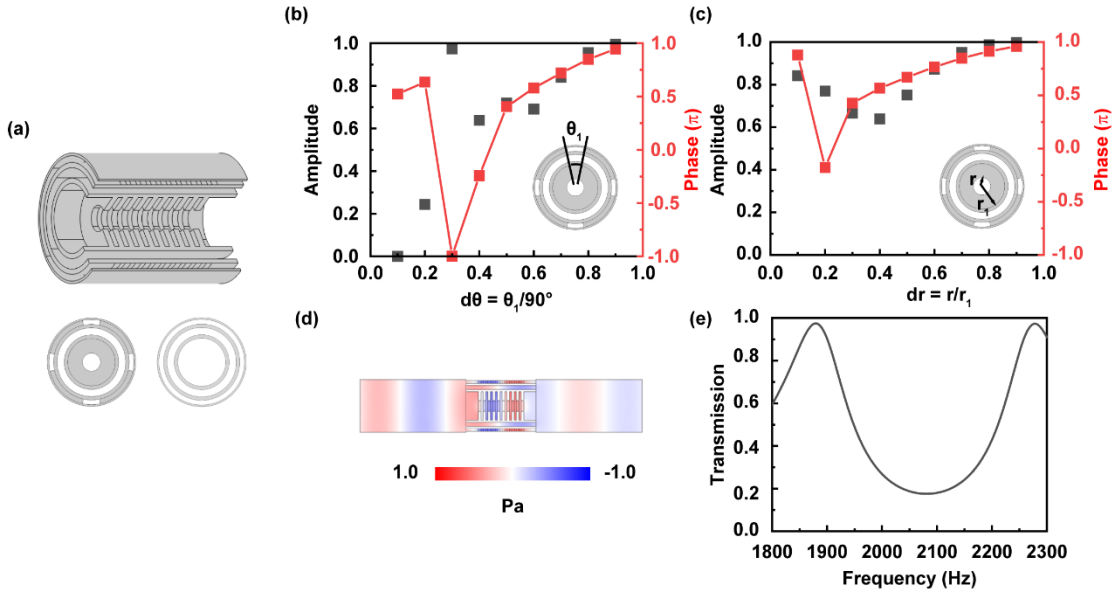


Fig. S5. Illustration and simulation results for cylindrical PGUOM (Design 2). (a) Illustration of the cylindrical PGUOM and its cross-section pattern. (b-c) Amplitude and phase of the transmission wave for the inner and outer layer at 2000 Hz, varying in angle and radius, respectively. (d) Pressure field distribution of the cylindrical PGUOM at 2000 Hz. (e) Transmission spectrum of the cylindrical PGUOM across the frequency ranging from 1800 Hz to 2300 Hz.

Supplementary Note 4: Limitation of the cylindrical PGUOM measurement setup

Fig. S6(a) illustrates the measurement setup for the *cylindrical PGUOM*, an impedance tube. Fig. S6(b-c) compares experimental results (red lines) and simulation results (black lines) for the abovementioned designs. Significant measurement errors become evident as the frequency exceeds 2000 Hz. To understand this phenomenon, we measured the transmission loss of an empty tube, as shown in Fig. S6(d). Ideally, the transmission loss should be consistently zero across all frequencies. However, the transmission loss becomes much more than zero when the frequency exceeds 2000 Hz. This observation strongly indicates that our measurement setup is insufficient when dealing with frequencies higher than 2000 Hz. As a result of this limitation, the target frequency is adjusted to 1500 Hz and the dimensions of the design are refined for the validation of the design.

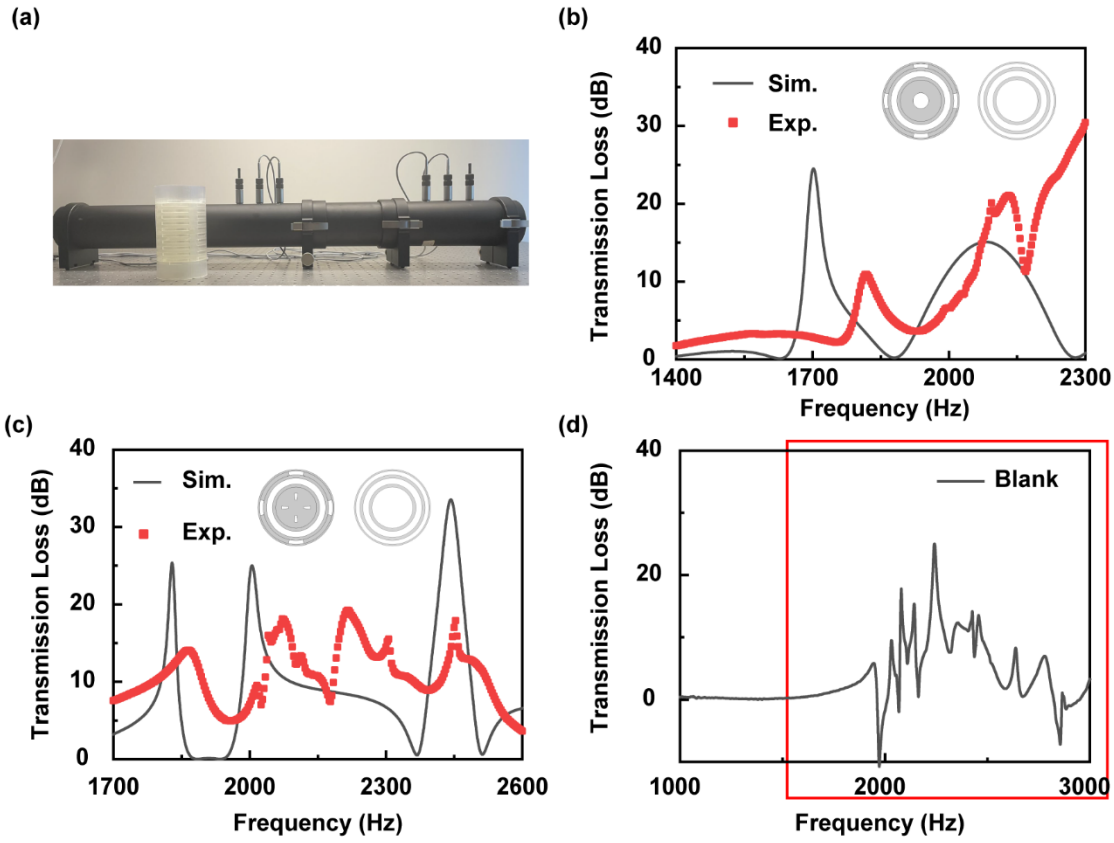


Fig. S6. Experimental setup and simulation results of the cylindrical PGUOM. (a) Illustration of the experimental setup and a sample of the cylindrical PGUOM. (b-c) Comparison of experimental results (red lines) and simulation results (black lines) for the transmission loss of Design 1 and Design 2, respectively. (d) Experimental results for the transmission loss of an empty tube shown in (a).

Supplementary Note 5: Acoustic properties of rectangular PGUOM around 2000 Hz and cylindrical PGUOM around 1500 Hz

We simulated the transmission spectra of the rectangular PGUOMs and cylindrical PGUOM at frequencies ranging from 1700 Hz to 2300 Hz and from 1000 Hz to 2000 Hz, respectively. The spectra show a relatively broad sound silencing band. The broadband sound silencing results from the special acoustic properties of PGUOM. Fig. S7 shows the transmission wave's amplitude and phase difference among unit cells of rectangular PGUOMs and cylindrical PGUOM around the target frequencies. For all designs, the transmission phase differences approximate 0.66π, and the amplitudes are acceptable around target frequencies.

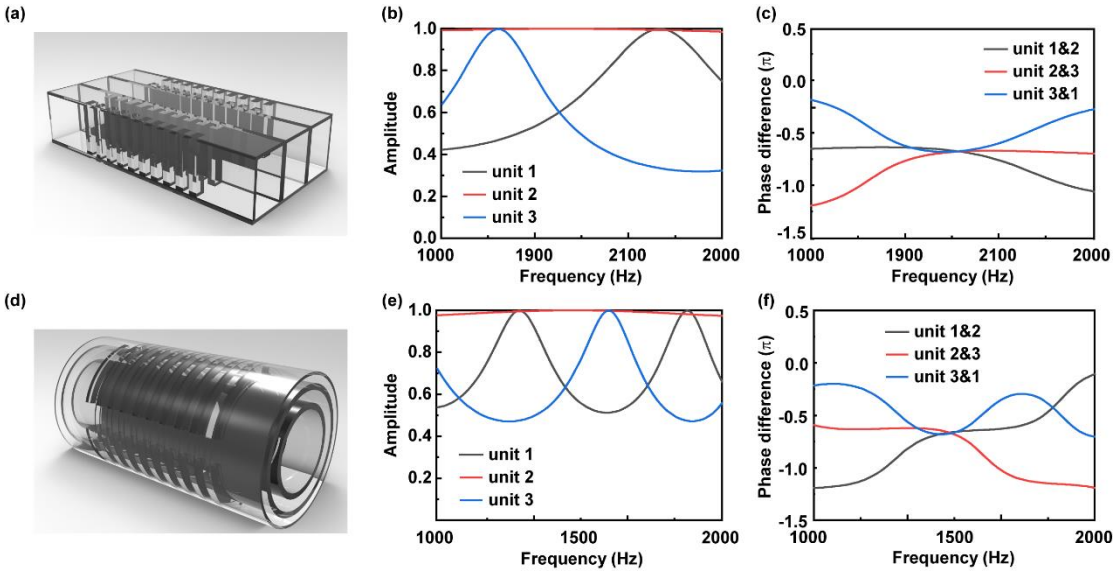


Fig. S7. Illustration and simulation results for rectangular and cylindrical PGUOM. (a) Illustration of the rectangular PGUOM. (b-c) Amplitude and phase differences among unit cell 1, unit cell 2 and unit cell 3 of rectangular PAUOM at frequencies ranging from 1700 Hz to 2300 Hz. (d) Illustration of the cylindrical PGUOM. (e, f) Amplitude and phase differences among unit cell 1, unit cell 2 and unit cell 3 of cylindrical PAUOM at frequencies ranging from 1000 Hz to 2000 Hz.

Supplementary Note 6: Thermoviscous losses effect on the transmission loss

Fig. S8 presents a comparison of the simulated transmission loss with and without the inclusion of thermoviscous losses, at different openness levels. Thermoviscous losses primarily arise within the narrow air channels of the structure, especially near rigid boundaries. To assess their contribution, we conducted simulations both with and without thermoviscous effects. The results show that thermoviscous losses reduce the transmission peak and increase the transmission loss of the dip, but they barely shift the center frequencies. This suggests that thermoviscous absorption plays a secondary role in the overall silencing effect and is not the dominant mechanism in the PGUOM design.

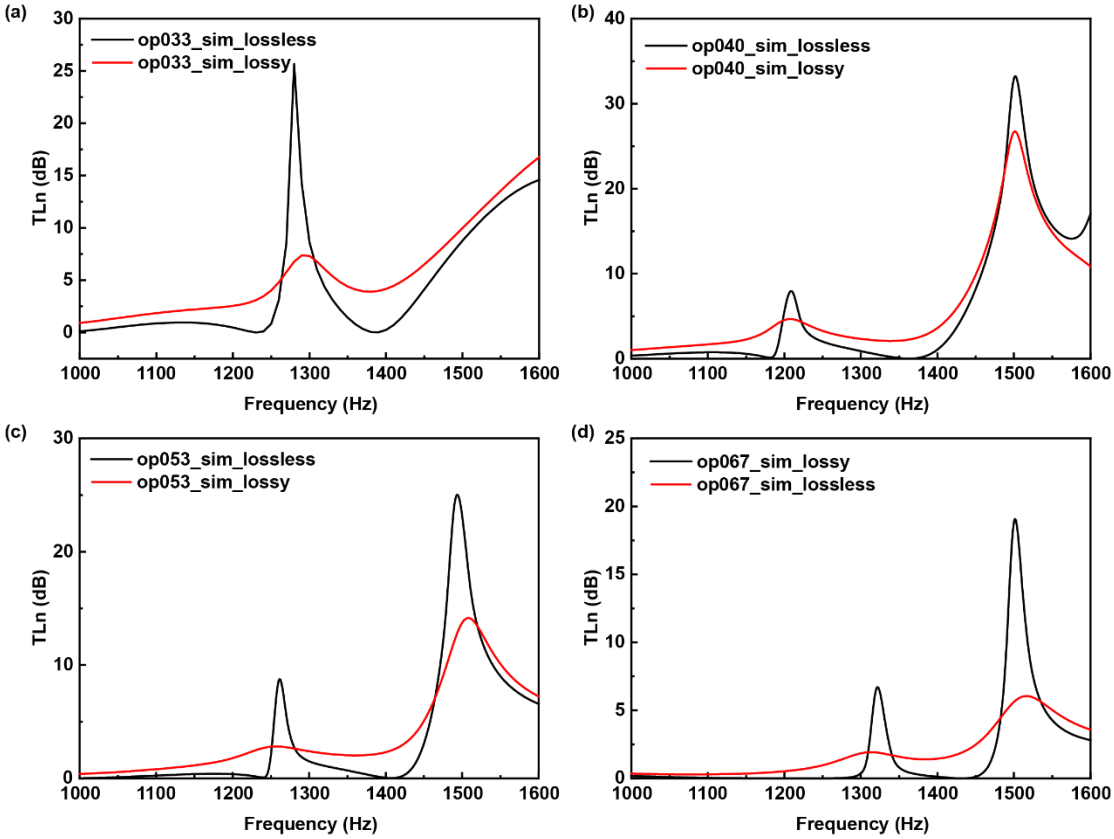


Fig. S8. Simulated transmission loss of the cylindrical PGUOM with and without thermoviscous losses at different openness levels.

Supplementary Note 7: Experimental setup and validation of rectangular PGUOM

In Fig. S9(a), we present an image of the experimental setup for measuring the rectangular PGUOMs. Two microphones are employed to record the incident and transmitted sound pressure. A movable stage allows precise positioning of the microphones for measurement. Additionally, two plates and resin walls or foams are utilized to minimize acoustic wave losses and prevent noise interference from outside, ensuring the accuracy of our measurement results. To maintain the consistency of our simulations with experimental data, the models we built in the simulations are identical to the experimental setup, as depicted in Fig. S9(b-c), representing the pressure field distributions with rigid boundary conditions or PML boundary conditions. Although we used point speaker instead of using planar speaker. The distributions clearly indicate that the incident wave has transformed from a point wave to a plane wave before interacting with the design. Furthermore, both designs effectively reduce the pressure of the transmission field to nearly zero after the sound passes through them.

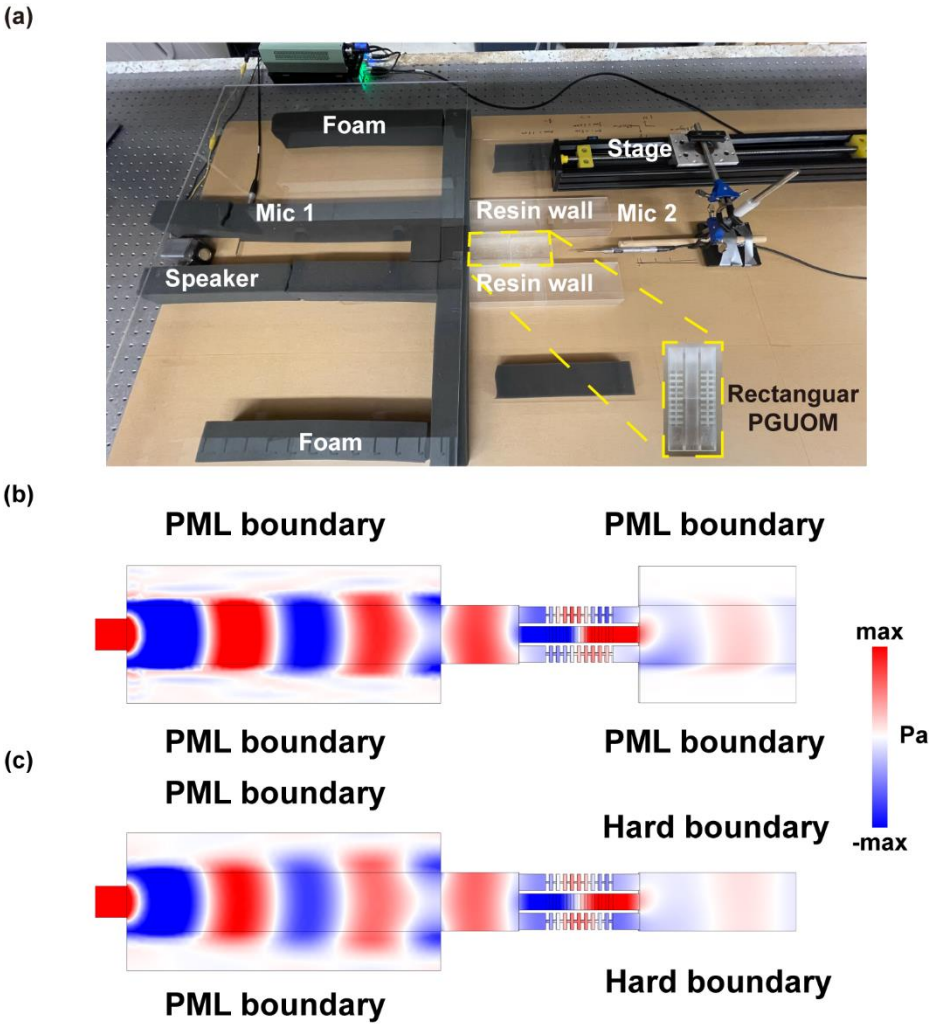


Fig. S9. Experimental setup and simulation results for rectangular PGUOM. (a) Experimental set up for the rectangular PGUOM. (b) Pressure field distribution of the PGUOM with PML boundary conditions. (c) Pressure field distribution of PGUOM with rigid boundary conditions.